

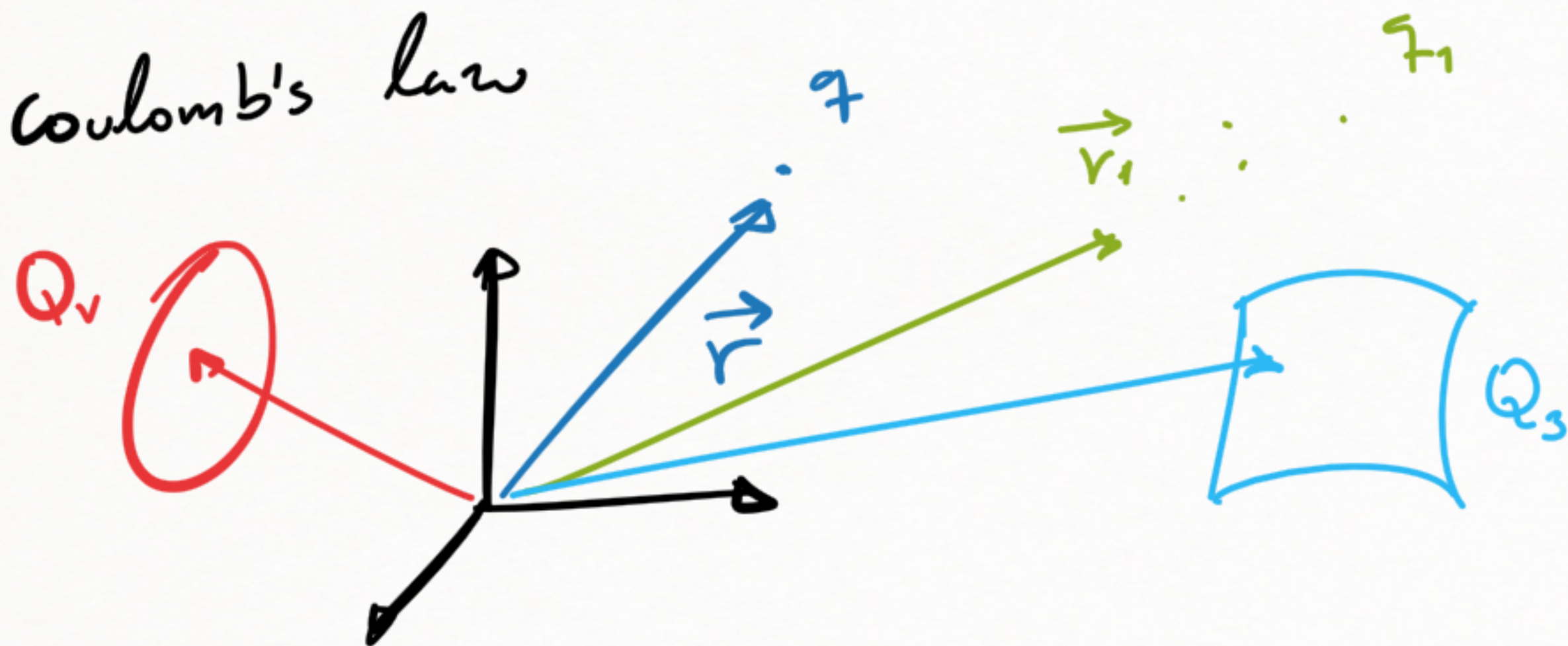
Electrodinámica

Clase #2

09/08/2025

Unit 1: Electrostatics

Coulomb's law



⇒ Electric field

Property of space due to the presence of electric charges

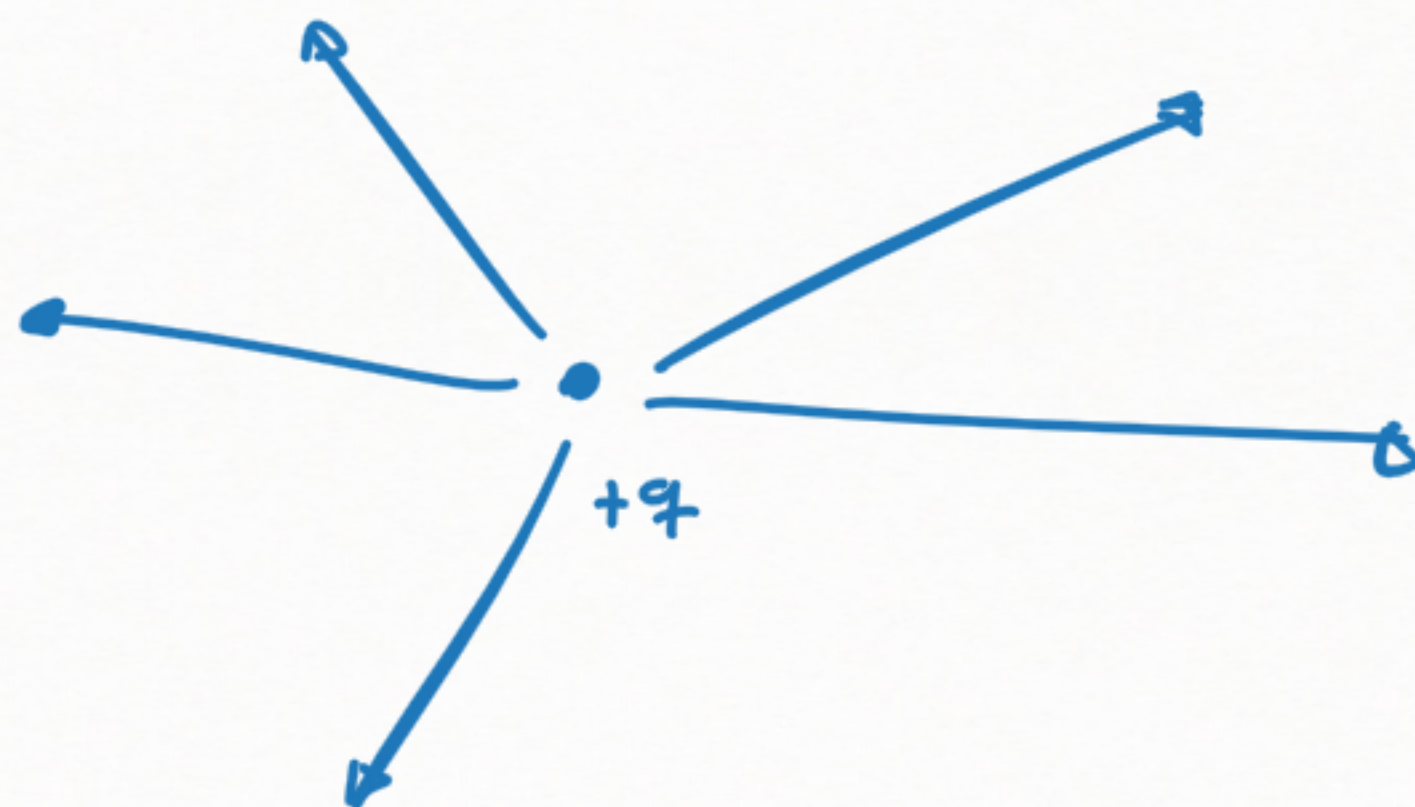
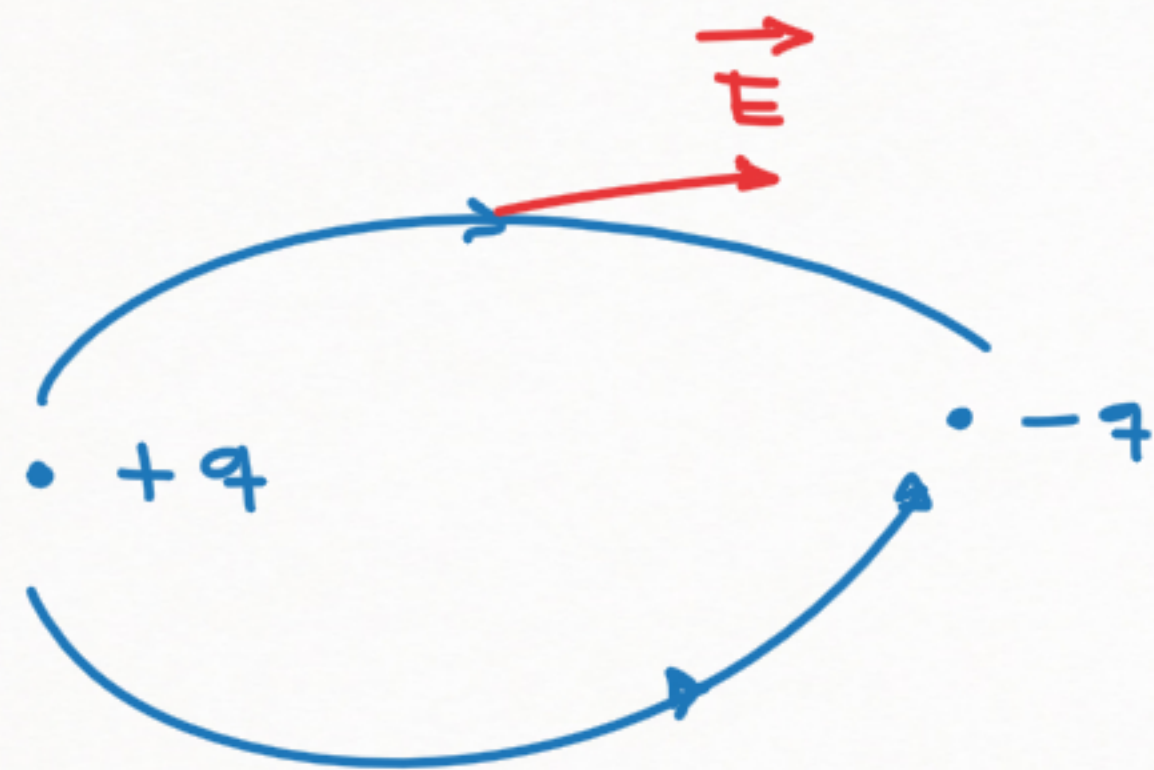
$$\vec{F}_q = \frac{q}{4\pi\epsilon_0} \sum_{i=1}^n \frac{q_i}{|\vec{r} - \vec{r}_i|^3} (\vec{r} - \vec{r}_i)$$

$$+ \frac{q}{4\pi\epsilon_0} \left[\int_S \frac{\sigma(\vec{r}')}{|\vec{r} - \vec{r}'|^3} d\alpha (\vec{r} - \vec{r}') \right. \\ \left. + \int_V \frac{\rho(\vec{r}'')}{|\vec{r} - \vec{r}''|^3} d\upsilon'' (\vec{r} - \vec{r}'') \right]$$

$$\vec{E} = \lim_{q_0 \rightarrow 0} \frac{F q_0}{q_0} = \frac{1}{4\pi\epsilon_0} \sum_{i=1}^n \frac{q_i}{|\vec{r} - \vec{r}_i|^3} (\vec{r} - \vec{r}_i) + \frac{1}{4\pi\epsilon_0} \int_S \frac{\sigma(\vec{r}') d\vec{s}'}{|\vec{r} - \vec{r}'|^3} (\vec{r} - \vec{r}') + \int_V \frac{\rho(\vec{r}'') dV''}{|\vec{r} - \vec{r}''|^3} (\vec{r} - \vec{r}'')$$

$\uparrow$
 $\vec{E}(r)$

$$\vec{E} = \mathbb{R}^3 \mapsto \mathbb{R}^3$$



Electrostatic Potential.

- Theorem: if $\vec{F}: U \rightarrow \mathbb{R}^3$ ($U \subset \mathbb{R}^3$) is a function defined on a simple connected set and $\nabla \times \vec{F} = 0$ in U then there exist $v: U \rightarrow \mathbb{R}$ such that.

$$\vec{F} = -\nabla v$$

$$\nabla \times \vec{F} = 0$$

$\Rightarrow$

$$\nabla \times \vec{E} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \partial_x & \partial_y & \partial_z \\ E_x & E_y & E_z \end{vmatrix}$$

mediant Levi-Civita

$$\{i\}_{i=1}^{n=3}$$

$$(\nabla \times \vec{E})_i = \epsilon_{ijk} \partial_j E_k$$

Levi-Civita
Symbol

$$\epsilon_{ijk} = \begin{cases} 0 & i=j \text{ or } i=k \text{ or } j=k \\ 1 & (i,j,k) \text{ is an even permutation of } (1,2,3) \\ -1 & (i,j,k) \text{ is an odd permutation of } (1,2,3) \end{cases}$$

Examples

i) $(1,1,3)$

ii) $(1,2,3)$

iii) $(2,1,3)$

$$\epsilon_{113} = 0$$

$$\epsilon_{123} = 1$$

$$\epsilon_{213} = -1$$

$\vec{E} = \vec{E}(\vec{r}) \rightarrow$ The electric field has a dependence
on $\vec{r}$ in the factor:

$$\frac{\vec{r} - \vec{r}'}{|\vec{r} - \vec{r}'|^3} = \frac{1}{|\vec{r} - \vec{r}'|^2} (\vec{r} - \vec{r}')$$

Note:

$$\nabla \times (f \vec{A}) = \nabla f \times \vec{A} + f \nabla \times \vec{A}$$

Proof:

Componente i -ésima:

$(\nabla \times (f \vec{A}))_i \rightarrow$ componente i -ésima del rot dado por $f \vec{A}$

Producto de
campo vectorial
 $\downarrow$

Tenemos

$$\hookrightarrow \epsilon_{ijk} \partial_j (f A_k)$$

$$\hookrightarrow \epsilon_{ijk} [\partial_j f A_k + f \partial_j A_k]$$

$$\vec{A} \times \vec{B}_i = \epsilon_{ijk} A_j B_k$$

$$\nabla f = \frac{\partial f}{\partial x} \hat{i} + \frac{\partial f}{\partial y} \hat{j} + \frac{\partial f}{\partial z} \hat{k}$$

Tenemos

$$\begin{aligned} \hookrightarrow (\nabla \times f \vec{A})_i &= \epsilon_{ijk} \overbrace{\partial_j f}^{\nabla f_j} A_k + f \epsilon_{ijk} \partial_j A_k \\ &\vdots \\ &= (\nabla f \times \vec{A})_i + f \nabla \times \vec{A}_i \end{aligned}$$

Tenemos

$$\nabla \times \left(\frac{1}{|\vec{r} - \vec{r}'|^3} (\vec{r} - \vec{r}') \right)$$

Use the product rule anterior:

$$= \nabla \left(\frac{1}{|\vec{r} - \vec{r}'|^3} \right) \times (\vec{r} - \vec{r}') + \frac{1}{|\vec{r} - \vec{r}'|^3} \underbrace{\nabla \times (\vec{r} - \vec{r}')}_{=0}$$

Proof

$$\nabla \times (\vec{r} - \vec{r}')_i = \epsilon_{ijk} \partial_j$$

$$\left. \begin{aligned} \vec{r} &= (x_1, x_2, x_3) \\ \vec{r}' &= (x'_1, x'_2, x'_3) \end{aligned} \right\}$$

$$\partial_j = \left(\frac{\partial}{\partial x_1}, \frac{\partial}{\partial x_2}, \frac{\partial}{\partial x_3} \right)$$

$$\left(\frac{\partial x_i}{\partial x_1} = 1 ; \quad \frac{\partial x_3}{\partial x_1} = 0 ; \quad \frac{\partial x_j}{\partial x_i} = \begin{cases} 1 & \text{si } i=j \\ 0 & \text{si } i \neq j \end{cases} \right)$$

*

$$\vdots = \epsilon_{ijk} \frac{\partial}{\partial x_j} (x_k - x'_k) = \epsilon_{ijk} \frac{\partial}{\partial x_j} x_k = \underbrace{\epsilon_{ijk} \delta_{jk}}_{\substack{\uparrow \\ \text{delta de Kronecker}}} = \epsilon_{ijj} = 0$$

• Note: contracciones

$$i) A_i \delta_{ij} = A_1 \delta_{1j} + A_2 \delta_{2j} + A_3 \delta_{3j}$$

$$\boxed{A_i \delta_{ij} = A_j}$$

$$ii) A_{ijk} \delta_{km} = A_{ijm}$$

Exercises:

1) $\nabla (\vec{A} \times \vec{B})_i$ ✓

$$\left[\nabla \left(\frac{1}{|\vec{r} - \vec{r}'|^3} \right) \right]_i = \partial_i \left(\frac{1}{\left[\sum_{j=1}^3 (x_j - x'_j)^2 \right]^{3/2}} \right)$$

2) $\nabla \times (\vec{F} + \vec{G})_i = \epsilon_{ijk} \partial_j (\vec{F} + \vec{G})_k$

$$\vdots = -3 \frac{x_i - x'_i}{|\vec{r} - \vec{r}'|^5}$$

3) $\left[\nabla \left(\frac{1}{|\vec{r} - \vec{r}'|} \right) \right]_i \hookrightarrow \epsilon_{ijk} [\partial_j F_k + \partial_j G_k]$

$$= \epsilon_{ijk} \partial_j F_k + \epsilon_{ijk} \partial_j G_k$$

Tenemos

$$\vdots = (\nabla \times \vec{F})_i + (\nabla \times \vec{G})_i$$

$$|\vec{r} - \vec{r}'| = \sqrt{(x-x')^2 + (y-y')^2 + (z-z')^2}$$

$$\vdots$$

$$= \sqrt{\sum_{i=1}^3 (x_i - x'_i)^2}$$

$$\vdots$$

$$= \left[\sum_{i=1}^3 (x_i - x'_i)^2 \right]^{1/2}$$

$$\nabla f = \frac{\partial f}{\partial x_1} \hat{i} + \frac{\partial f}{\partial x_2} \hat{j} + \frac{\partial f}{\partial x_3} \hat{k} \quad ; \quad \nabla f_i = \frac{\partial f}{\partial x_i} = \partial_i f$$

Tenemos

$$\frac{\partial}{\partial x_i} \left(|\vec{r} - \vec{r}'| \right) = \nabla_i |\vec{r} - \vec{r}'|$$

Derivamos

$$= \partial_i \left(\left[\sum_{j=1}^3 (x_j - x_j')^2 \right]^{1/2} \right)$$

Chain rule

$$A_j \delta_{ij} \rightarrow A_i$$

$$\vdots$$
$$= \frac{1}{2} \left(\sum_{j=1}^3 (x_j - x_j')^2 \right)^{-1/2} \cdot 2 \sum_{j=1}^3 (x_j - x_j') \cdot \frac{\partial x_j}{\partial x_i} =$$

$$\frac{1}{|\vec{r} - \vec{r}'|} (x_i - x_i') \quad \therefore \quad \nabla |\vec{r} - \vec{r}'| = \frac{\vec{r} - \vec{r}'}{|\vec{r} - \vec{r}'|^2}$$

Ejercicio 3

$$\partial_i \left(\sum_{j=1}^3 (x_j - x_j')^2 \right)^{-1/2} = -\frac{1}{2} \left(\sum_{j=1}^3 (x_j - x_j')^2 \right)^{-3/2} \cdot 2 \sum_{j=1}^3 \overbrace{(x_j - x_j')}^{A_i} \overbrace{\delta_{ij}}^{\delta_{ij}}$$

$$\Rightarrow \frac{1}{|\vec{r} - \vec{r}'|^3} (x_i - x_i') ; \quad \text{tenemos}$$

$$\nabla \left(\frac{1}{|\vec{r} - \vec{r}'|} \right) = - \frac{\vec{r} - \vec{r}'}{|\vec{r} - \vec{r}'|^3}$$